



LUNDS UNIVERSITET

Design of Experiments

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Project

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1 Introduction

Sodas are widely consumed all around the world, and have become a large part of global culture. Several big name brands might come to mind even at the mention of the word soda, but can people actually tell the difference compared to a generic knock-off brand? Additionally, in a "health-washing" effort, almost all brands have formulated so-called "diet" versions of their sodas, using artificial sweeteners instead of refined sugars. In this experiment though, we are solely interested in the taste perceptions, attempting to answer the question whether or not people can tell the difference between diet and regular versions of the same brand soda.

Since different brands might use different kinds of artificial sweeteners, it is very possible that there might be a large interaction effect between the brand and diet version variables. As such, utilizing a factorial design seems like the best strategy to tackle the problem. Even if we had been interested in only one of the individual variables, for example to see if people generally like regular or diet versions better, simply blocking for brand would have been inadequate, as blocking assumes the variables to be independent.

It is a well-known fact that temperature can affect taste perceptions to a large degree - as an extreme example, consider a room-temperature, melted ice cream, which can appear almost sickly sweet, perceived to have a perfect level of sweetness if consumed appropriately cold. Likewise, one could expect that the temperature of soda could affect how people perceive its taste. As such, one could consider adding temperature to the model, either as a factor in its own right, or as a blocking variable, if no interaction effects are expected. However, much as with ice cream, sodas are most commonly served straight from a refrigerator, so we deemed it a factor of no interest. By keeping the bottles of soda in the fridge in between runs of the experiment, and only pouring right before serving, we could also keep the temperature approximately constant between each replicate. Of course, this means that our results cannot be generalized to other temperatures, but because of aforementioned reasons, we deemed this to be of little or no interest.

We thus arrived at a two-factor full factorial design, and decided to test three brands of sodas (CocaCola, Pepsi, CubaCola), as well as regular vs diet version (CocaCola Zero, PepsiMax, CubaCola Zero), so a 3x2 design. We have the effects model as follows:

$$Y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta_{ij}) + \epsilon_{ijk}$$

where Y is the outcome variable, μ is the overall mean, τ the three-level brand factor, β the two-level diet factor, $\tau\beta$ their interaction, and ϵ the random error, assumed to be distributed $NID(0, \sigma^2)$. In the hopes of getting somewhat close to a normal distribution, while being restricted for monetary reasons, we chose to perform 25 replicates, i.e., gather 25 volunteers for the soda testing.

As for choice of outcome variable, we considered a few different options, mainly between a relative scale, participants ranking the sodas from 1-6 after testing all of them, vs an absolute scale, ranking the sodas immediately after tasting each one from 1-10. Both scales would be for taste preference. Ultimately, we settled on the latter option, as we deemed a relative ranking more prone to error and bias. For instance, we argued that someone might, having tested all sodas, try to second-guess their opinions of the sodas, trying to figure out which brand was which, and potentially adjusting to their preconceived presumed

preferences. There is also the danger that the order in which sodas were tested could play a greater impact if trying to rank them at the end, remembering later ones more vividly for example. As such, having the subjects rate immediately after tasting was deemed the potentially safest route that would avoid as many lurking variables as possible. Given that some subjects might like sodas as a whole more than others, we also kept in mind that as a potential fix after gathering the data, we could re-scale the outcome variables, a decision that was left until after having gathered the data.

As the taste of a different soda could potentially linger in the mouth between runs (unique combinations of brand and diet/regular, that each participant had to taste test all 6 possible of), we required participants to have a palate cleanser with water in between each run, to minimize this potential influence of run order. Since a large part of the experience of drinking sodas is the mouthfeel and actually drinking the soda, we decided to give each participant 50 ml of each soda to test (as opposed to for instance a wine tasting, where participants would spit out the wine). We also figured that it would be potentially easier to get willing participants if they were allowed to swallow, in addition to the benefit of more closely mimicking the usual soda experience.

After gathering the data as above, we will test the three null hypotheses

$$H_{01} : \tau_1 = \tau_2 = \tau_3 = 0$$

$$H_{02} : \beta_1 = \beta_2 = 0$$

$$H_{03} : (\tau\beta)_{11} = \dots = (\tau\beta)_{32} = 0$$

using standard ANOVA with F-statistics. In addition, we will construct graphs to help with interpretation, as well as attempt to fit a linear model to our data.

2 Regression Model

Next we construct a linear model to study in greater detail how brand and diet affect the scores of the soda. We construct a linear model by setting score against diet, brand, and their interactions:

$$\text{Score} = \beta_0 + \beta_1 \text{Regular} + \beta_2 \text{Generic} + \beta_3 \text{Pepsi} + \beta_4 \text{Generic*Regular} + \beta_5 \text{Pepsi*Regular} + \epsilon$$

Beta	Slope	Confidence Interval
β_0	4.92	[4.19, 5.64]
β_1	-0.0367	[-1.05, 0.975]
β_2	-0.0320	[-1.03, 0.970]
β_3	0.843	[-0.168, 1.85]
β_4	0.352	[-1.06, 1.76]
β_5	0.277	[-1.15, 1.70]

Table 1: Slope and Confidence Intervals for Coefficients

Here we can see that the average score given was around a 5 which one would expect, and that the most significant factor seems to be Pepsi on its own. We will study this further by plotting the residuals on a QQ-Normal Plot, and then plotting a box plot of the residuals by brand and by diet.

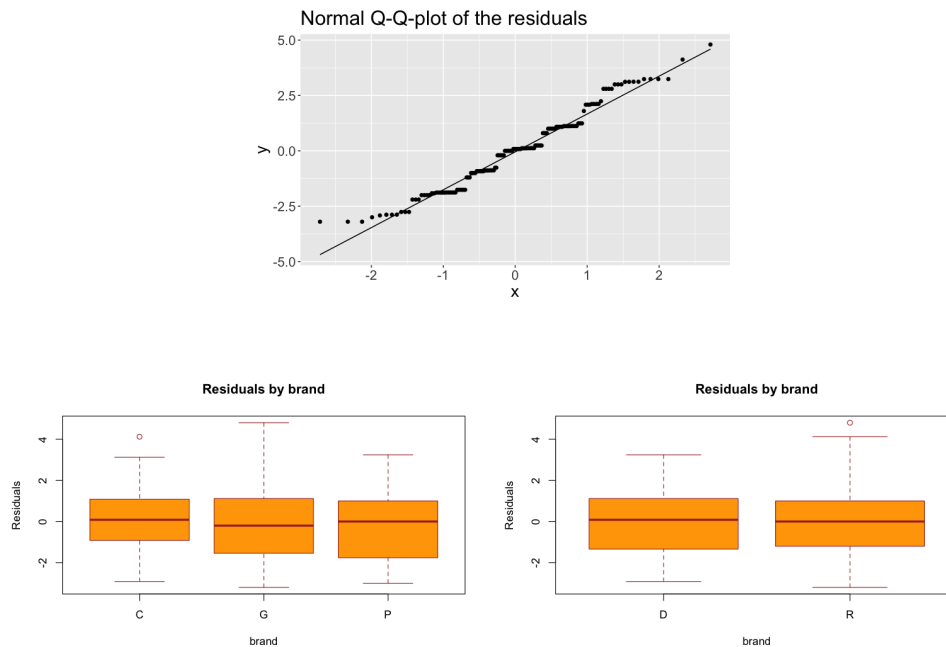


Figure 2: Residuals for Linear Model

Looking at these figures we can see that the residuals are fairly normally distributed. There are some slightly larger than average residuals for our Generic brand but otherwise everything looks as it should.

Now that we know that our data is normally distributed we can plot the scores against brand and diet to see what factors are most relevant. We will do this by creating a box plot of the scores separated first by the type of soda, and then by if they are diet or regular. These box plots will also be equipped with the average score, confidence intervals, and prediction intervals in order to fully understand how each variable effects the score.

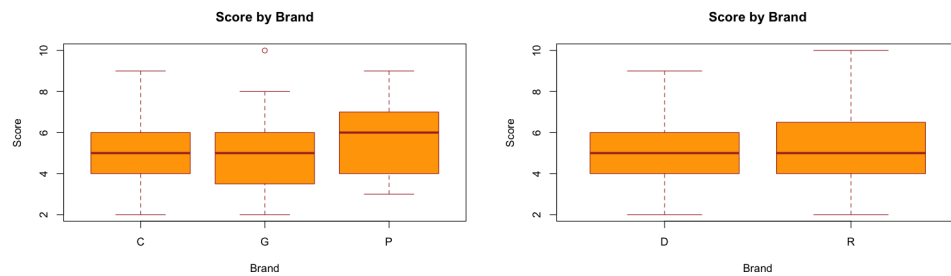


Figure 3: Score Separated by Variable

Here we can see that the CocaCola and Generic brands are almost identical in how their scores are distributed, with the Generic brand average only being slightly higher than Coke's. However, we can also see that Pepsi has what looks like a significantly higher average score. We can also see from our second plot that Diet and Regular sodas have almost the exact same box plot. The only discernible difference between these two plots being that the Diet soda has a slightly smaller confidence interval. We will test these factors further by constructing an ANOVA table for the factors to see which variables influence the score given by the participants.

3 ANOVA analysis

As we are interested into looking at the effect of 2 independent variables (Brand and Diet version) that could possibly affect our response variable, hence, we decided to use factorial design and conduct the ANOVA analysis. Our experiment is constituted of two factors with different number of levels, Brand has 3 levels and Diet has 2 levels.

Firstly, here is a table of processed data, please note that the values inside are averages of the observations (participants scores) for specific factor combinations.

	Brand		
	CocaCola	Pepsi	CubaCola
Diet	5,08	5,76	4,80
Not Diet	4,80	6,00	5,20

Figure 4: Averages of scores for various treatment combinations

From this data we can see that all brand combinations have received similar responses with highest being non-diet version of Pepsi (6,00) and lowest being diet CubaCola and non-diet version of CocaCola. Additionally, we can see that both versions of Pepsi performed the best and the reason for it could be in the type of sweetener that company uses or the carbonation level that company uses. If we look at the Raw Data Table (in the appendix) we can notice that as the experiment ran on, performers of the experiment rated different treatments mostly between 4 and 6. The reason for this could be that as the time went on, samples became less carbonated and therefore, tasted more similarly.

Now, proceeding to observe the results of ANOVA analysis. Here is the completed ANOVA table, result of the code in RStudio (code will be in the appendix).

```

      Df Sum Sq Mean Sq F value Pr(>F)
Brand    2   28.1   14.051    4.382 0.0142 *
Diet     1    1.2    1.152    0.359 0.5499
Brand:Diet 2    0.9    0.426    0.133 0.8756
Residuals 144 461.7    3.206

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> |

```

Figure 5: ANOVA Table for preference scores

From the ANOVA table, we could conclude that only the brand effect reached statistical significance. As such, it is unlikely that the brand name did not effect the taste perception, and people really could tell a difference between different soda brands - however, they could not distinguish whether they were diet versions or not. Additionally, the interaction effect seems to also have been non-existent.

Here we have created the interaction plot to help us visualize the results.

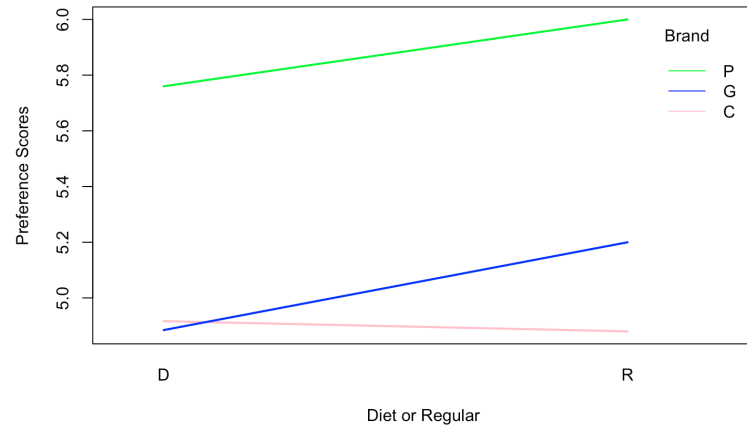


Figure 6: Brand-Diet interaction plot

As seen in the plot, the significant factor is indicated by the lack of parallelism of the lines. In general, higher average scores are attained for regular versions of soda, for Pepsi and CubaCola. Changing from diet to regular version, average scores for CocaCola may actually decrease, whereas they increase for Pepsi and CubaCola. Therefore, Pepsi seems to give the best results if we want high average scores for both diet and regular soda.

4 Conclusion

There are several interesting things to consider here. Firstly, looking at the ingredients for all three different brands, we can see that their respective Diet versions all use the same two artificial sweeteners, aspartame and acesulfame K. Thus it is perhaps no wonder that we see no interaction effects, as the change of ingredients was the same. On the other hand, potentially, they could still have used different amounts and proportions of the sweeteners, and there could have been some interaction with the rest of the recipes, such that they would have harmonized better or worse with one or the other.

That said, as the Diet effect itself was not significant, we cannot reject the null hypothesis that there was no difference in taste perception between the Regular and Diet versions to begin with. Given this, that the people in our study preferred the Diet and Regular versions just as much, it seems that the artificial sweeteners in general performed as well as regular sugar - and then the only real reason why there would be an interaction effect with Brand would have been that one of the brands would have made a mistake in their formulation, which seems quite unlikely.

Does this mean that people couldn't tell the difference between Diet and Regular versions? Not necessarily. First of all, albeit we got a very high p-value, all that means is that our experiment failed to reject that there was a Diet version effect, so it might be a limitation of our experiment. Even if we had, for example, had a larger sample size of people, we only asked the participants to rate the sodas to their taste preference - as such, maybe they could tell a difference, but just liked the different versions about as much on the abstract 1-10 scale. As such, it could be interesting to also include some sort of qualitative measure in further studies, where participants are invited to describe the flavours and what they like about each soda directly after tasting them.

Furthermore, it is also possible, that some people in our study preferred Diet versions to the Regular, whereas others preferred Regular versions, thus evening out the scores overall, rather than that individuals liked both versions about the same. All we have studied is some kind of average for each category, across lots of people. For this reason, one could consider perhaps using blocking for each participant, if one was interested in seeing whether or not individuals can tell the difference between Diet and Regular versions. However, it could be argued that the average values are in the end the more interesting ones, as that says more about the general palatability of Diet versions as compared to the Regular, which is what we would be interested in for product development reasons.

Given that we found no significant effect from Diet version of the sodas, it would seem that brands could safely replace regular sugar with these two artificial sweeteners, without affecting how much people like their products. At least, for sodas - it is not unreasonable to assume that the taste could still be affected in different kinds of products, such as baked goods, which would need further studies. Also, as ours was a relatively small study, it would be beneficial to try to replicate the results, while maybe also trying to block or otherwise control for other factors. For instance, as taste preferences develop over time, someone who is more used to artificially sweetened beverages, might well have a greater preferences for them than someone who is drinking one for the first time in the experiment. Although a large random sample of participants might help with this, it could be interesting to at least gather some additional data on the

participants themselves, like previous exposure, as well as general soda-drinking habits, as well as perhaps self-reported preferences. Especially given the very large residual sum of squares, it would seem that we have not explained a lot of the variance in taste perception at all, so further factors to include and/or block for would appear advisable.

Of course, our study only considered the taste of the sodas. It did not consider whether or not it would be beneficial to use artificial sweeteners, in general and the two used by the three brands in this study in particular, over regular sugar in the first place, which requires separate studies into health measures and outcomes. However, if there is some benefit, or less detriment, to using the artificial sweeteners, it would seem that the switch could be done without sacrificing taste. Or at least how people in general would perceive it, which might further be limited to the specific demographic of our study participants.

Finally then, we consider the brand effect, which did reach statistical significance on significance level 0.05. As such, it seems that our study participants overall did prefer the tastes of the different brands to differing degrees. Given the relatively small number of participants, however, we might have simply happened across a group of people all with a preference for the tastes of one specific soda - which, on the other hand, would be interesting in itself, that they then actually could tell the difference. Something that could be interesting for further studies, would be, aside from potentially increasing the number of participants, and trying to broaden the reach (rather than collecting the data at one isolated time and space), would be to also gather data on self-reported brand preferences beforehand. It would, for instance, be interesting to see if the people who said that they liked CocaCola/CubaCola/Pepsi the best, actually rated the respective brand higher, on average. It might be harder to include it as an actual factor in the ANOVA, since there might also be people proclaiming no preference, which would have to be thought of how to handle.

5 Appendix

Here is the raw data of the observations:

1	E 5	A 3	F 9	D 2	B 8	C 9
2	A 6	C 7	E 5	D 6	F 9	B 8
3	C 3	F 4	A 2	E 6	D 8	B 5
4	E 5	F 6	D 3	A 8	C 6	B 7
5	C 6	A 5	B 6	E 5	F 7	D 3
6	F 5	C 5	E 6	B 3	A 4	D 4
7	F 4	E 7	C 6	B 3	D 5	A 5
8	B 4	A 6	F 6	C 7	E 4	D 8
9	C 6	E 4	A 3	F 6	D 4	B 6
10	A 5	C 4	D 4	E 5	B 5	F 5
11	F 3	B 2	A 4	D 3	E 5	C 4
12	E 5	C 7	B 2	A 5	F 6	D 5
13	A 8	E 6	D 7	F 9	B 10	C 7
14	F 4	E 4	D 5	B 6	C 6	A 4
15	C 5	B 5	A 3	D 3	E 4	F 4
16	B 4	A 3	C 4	D 6	E 4	F 3
17	B 5	C 6	A 4	D 5	E 7	F 6
18	D 4	A 3	F 4	B 4	E 2	C 4
19	E 4	A 3	C 5	F 3	D 2	B 2
20	A 7	D 6	E 6	B 8	C 7	F 7
21	A 6	B 6	F 6	D 6	C 6	E 5
22	A 6	E 6	B 5	F 7	C 7	D 5
23	E 3	C 5	F 4	B 3	D 3	A 3
24	E 7	B 8	F 9	C 9	A 9	D 7
25	F 8	A 5	E 7	D 6	B 5	C 9

Figure 7: Raw data table of the observations obtained in random order

Here is the R code used to create the linear model and the plots:

```

7 = ```{r}
8 library("tidyverse")
9 data <- read.csv("experiment - Sheet1.csv")
10 model <- lm(score ~ diet*brand, data= data)
11 model$coefficients
12 confint(model, level = 0.95)
13 score.pred <-
14   cbind(data,
15     fit = predict(model),
16     conf = predict(model, interval = "confidence"),
17     pred = predict(model, interval = "prediction"))
18 score.predsconf.fit <- score.predsconf.fit <- NULL
19 score.predsfit <- model$residuals
20 max.e <- max(abs(score.predsfit))
21 score.lim.e <- c(-max.e, max.e)
22
23 = ```

25 = ```{r}
26
27 ggplot(data = score.pred, aes(sample = e)) +
28   geom_qq() + geom_qq_line() +
29   labs(title = "Normal Q-Q plot of the residuals") +
30   theme(text = element_text(size = 18))
31
32 data$brand <- as.factor(data$brand)
33 boxplot(e ~ brand, data=score.pred,
34   main="Residuals by brand",
35   xlab="brand",
36   ylab="Residuals",
37   col="orange",
38   border="brown")
39   labs(caption = "original scale")
40
41 boxplot(score ~ brand, data=score.pred,
42   main="Score by Brand",
43   xlab="Brand",
44   ylab="Score",
45   col="orange",
46   border="brown")
47   labs(caption = "original scale")
48 = ```

50 = ```{r}
51 data$brand <- as.factor(data$brand)
52 boxplot(e ~ diet, data=score.pred,
53   main="Residuals by brand",
54   xlab="brand",
55   ylab="Residuals",
56   col="orange",
57   border="brown")
58   labs(caption = "original scale")
59
60 boxplot(score ~ diet, data=score.pred,
61   main="Score by Brand",
62   xlab="Brand",
63   ylab="Score",
64   col="orange",
65   border="brown")
66   labs(caption = "original scale")
67 = ```

```

Figure 9: R code Linear Model

R code for ANOVA table, Interaction plot and the residuals:

```

1 |
2
3 data = read.csv("experimentproba1.csv")
4 attach(data)
5 av = aov(Score ~ Brand*Diet)
6 summary(av)
7 detach(data)
8
9 interaction.plot(x.factor = data$Diet, #x-axis variable
10   trace.factor = data$Brand, #variable for lines
11   response = data$Score, #y-axis variable
12   fun = mean, #metric to plot
13   ylab = "Preference Scores",
14   xlab = "Diet or Regular",
15   col = c("pink", "blue", "green"),
16   lty = 1, #line type
17   lwd = 2, #line width
18   trace.label = "Brand")
19
20 qqnorm(residuals(av))
21 qqline(residuals(av))
22

```

Figure 10: R code